

Standalone numerical study for a positive-rate square-root gDMR parameter set

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1 Numerical studies

This standalone document repeats the manuscript numerical-study design for the square-root gDMR parameter set reported in Table 1, using the positive risk-free rate $r = 0.05$. The option is a Bermudan put with twelve exercise dates, maturity $T = 1$, and strikes $K = 70, 80, 90, 100, 110$. The direct plain LSMC estimator is compared with the Hybrid LSMC–PDE estimator. All prices, standard errors, confidence intervals, and relative errors are computed from the direct estimators, and benchmark-relative errors use the plain LSMC references in Table 2.

The benchmark references are obtained with 1200 Euler time steps and 1, 200, 000 direct paths. Across the five strikes, the benchmark direct prices range from 3.734195 to 15.732048. These values define the reference scale used in the step-sweep and path-sweep experiments below.

1.1 Fixed path budget, varying number of Euler time steps

The first experiment keeps the direct path budget fixed and varies the number of Euler time steps. The two path budgets are 20, 000 and 60, 000, and both methods are evaluated at 24, 48, 72, and 96 steps. This isolates how the time discretization affects the direct Bermudan estimator away from the benchmark regime.

For 20, 000 paths, the LSMC relative-error range over the full step-sweep grid is 10.292%–41.074%, while the Hybrid LSMC–PDE range is 9.487%–28.662%. Figure 1 shows the strike-wise relative-error profiles, and Table 3 reports the representative 72-step values.

For 60, 000 paths, the same design gives an LSMC relative-error range of 10.731%–35.979% and a Hybrid LSMC–PDE range of 9.557%–29.642%. Figure 2 and Table 4 give the corresponding graphical and tabular summaries.

Table 1 Model and numerical parameters for the standalone positive-rate square-root study.

Quantity	Value
Model	gDMR
Option	Bermudan put
S_0	100.0
K	70, 80, 90, 100, 110
T	1.0
r	0.05
v_0	0.114
v'_0	0.09
κ_1	2.0
κ_2	0.1
θ	0.09
ξ_1	4.0
ξ_2	0.502
ρ_{12}	-0.3
ρ_{13}	-0.727
ρ_{23}	0.590
δ_1	0.5
δ_2	0.5
Exercise dates	12
Benchmark	LSMC, 1200 steps, 1,200,000 paths
Hybrid asset grid	301

Table 2 Benchmark direct LSMC reference prices for the standalone study.

Strike	Price	SE	CI lower	CI upper
70	3.734195	0.010500	3.713615	3.754775
80	5.384271	0.012928	5.358932	5.409610
90	7.683607	0.015509	7.653209	7.714005
100	10.959943	0.018058	10.924549	10.995337
110	15.732048	0.019990	15.692868	15.771228

1.2 Fixed number of Euler time steps, varying path counts

The second experiment fixes the number of Euler time steps and varies the direct path budget. The reported path grid is 250, 1000, 5000, 10000, 20000, 40000, and 60000. The two discretization levels are 48 and 60 steps.

At 48 steps for the at-the-money strike, the LSMC relative-error range over all path counts is 19.633%–49.645%, while the Hybrid LSMC–PDE range is 15.300%–19.872%. Figure 3 reports the full strike-wise path sweep, and Table 5 reports the 20,000-path representative values.

At 60 steps for the at-the-money strike, the corresponding LSMC range is 18.367%–55.332%, and the Hybrid LSMC–PDE range is 14.133%–19.266%. Figure 4 and Table 6 provide the companion summaries.

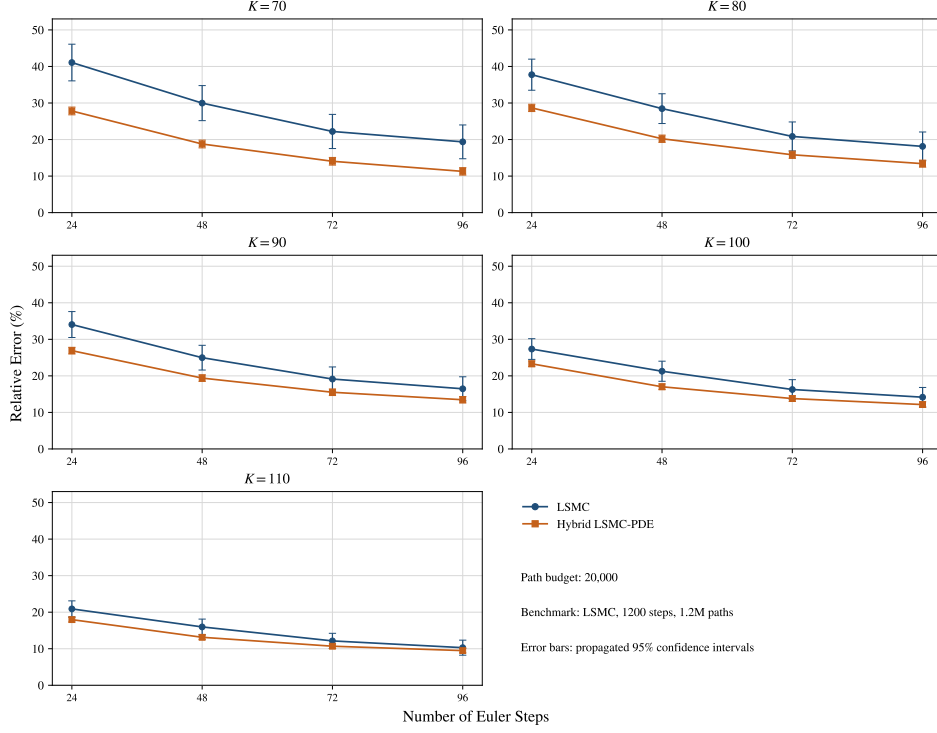


Fig. 1 Relative errors for the matched 20,000-path step-sweep experiment. Error bars are induced by the 95% confidence intervals of the direct estimators.

Table 3 Representative direct metrics at 20,000 paths and 72 Euler steps.

Strike	LSMC price	LSMC rel. err.	Hybrid price	Hybrid rel. err.
70	4.563583	22.211%	4.258975	14.053%
80	6.506968	20.851%	6.236673	15.831%
90	9.153437	19.129%	8.875658	15.514%
100	12.743236	16.271%	12.471011	13.787%
110	17.642070	12.141%	17.413834	10.690%

Table 4 Representative direct metrics at 60,000 paths and 72 Euler steps.

Strike	LSMC price	LSMC rel. err.	Hybrid price	Hybrid rel. err.
70	4.597115	23.109%	4.281651	14.661%
80	6.559007	21.818%	6.281833	16.670%
90	9.199141	19.724%	8.947173	16.445%
100	12.790717	16.704%	12.565771	14.652%
110	17.665909	12.292%	17.519654	11.363%

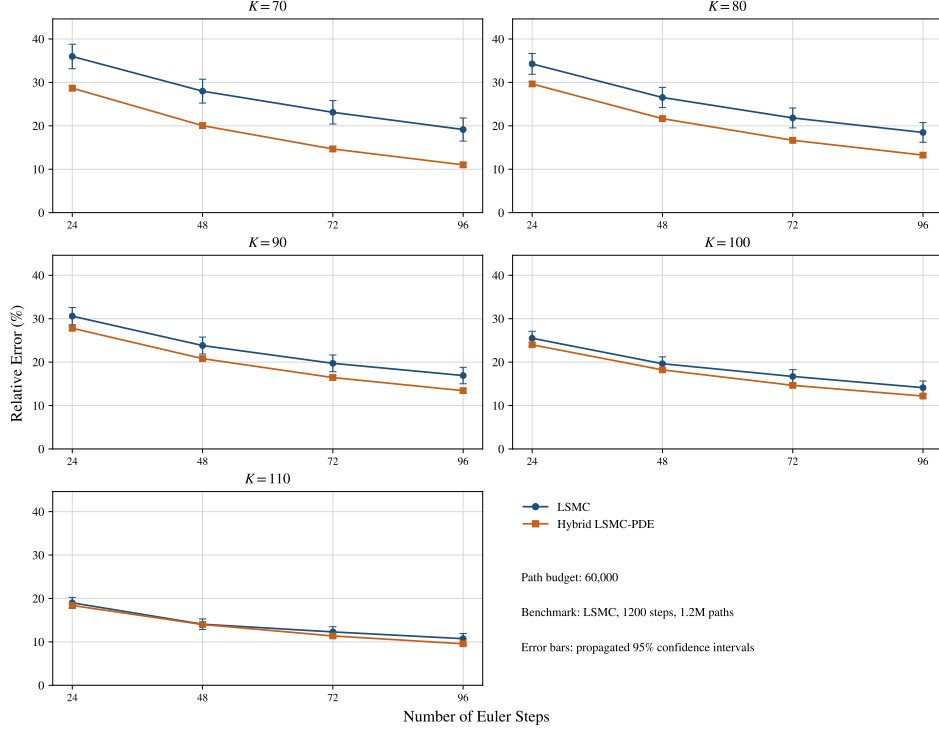


Fig. 2 Relative errors for the matched 60,000-path step-sweep experiment. Error bars are induced by the 95% confidence intervals of the direct estimators.

Table 5 Representative direct metrics at 20,000 paths under the fixed 48-step path sweep.

Strike	LSMC price	LSMC rel. err.	Hybrid price	Hybrid rel. err.
70	4.853248	29.968%	4.435483	18.780%
80	6.916490	28.457%	6.472788	20.217%
90	9.602840	24.978%	9.173708	19.393%
100	13.290796	21.267%	12.825020	17.017%
110	18.244102	15.968%	17.798200	13.133%

Table 6 Representative direct metrics at 20,000 paths under the fixed 60-step path sweep.

Strike	LSMC price	LSMC rel. err.	Hybrid price	Hybrid rel. err.
70	4.754932	27.335%	4.331094	15.985%
80	6.771642	25.767%	6.331029	17.584%
90	9.485072	23.446%	8.991973	17.028%
100	13.075526	19.303%	12.606509	15.023%
110	17.986559	14.331%	17.559868	11.618%

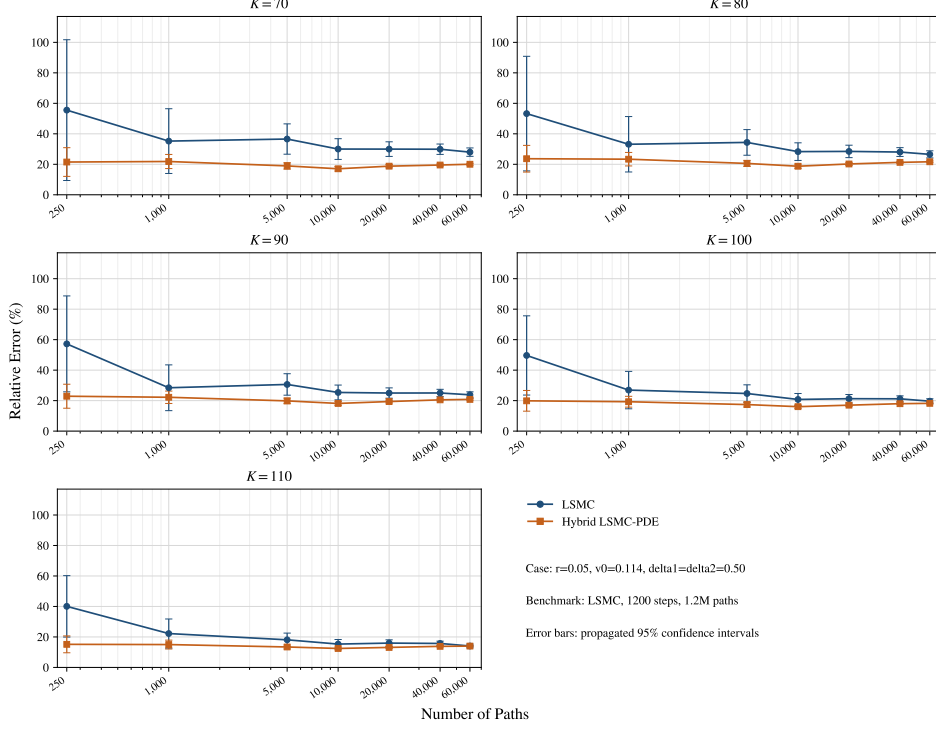


Fig. 3 Relative errors for the fixed 48-step path-sweep experiment. Error bars are induced by the 95% confidence intervals of the direct estimators.

1.3 High-path Hybrid confirmation

The fixed-step sweeps show that the coarse 48- and 60-step designs do not by themselves remove the discretization and grid effects for this square-root case. As a final confirmation, the Hybrid LSMC–PDE calculation is repeated with 1200 Euler steps, 100,000 direct paths, and a widened 601-point asset grid on $[0.10S_0, 5S_0]$. Under this accuracy-oriented setting, the largest benchmark-relative Hybrid error over the five strikes is 0.837%, and each strike is below the one-percent level.

Table 7 High-accuracy Hybrid LSMC–PDE prices with 1200 steps, 100,000 paths, and a widened 601-point asset grid.

Strike	Benchmark	Hybrid price	SE	Rel. error
70	3.734195	3.756622	0.008778	0.601%
80	5.384271	5.407195	0.011597	0.426%
90	7.683607	7.705869	0.014810	0.290%
100	10.959943	11.012766	0.018194	0.482%
110	15.732048	15.863678	0.021166	0.837%

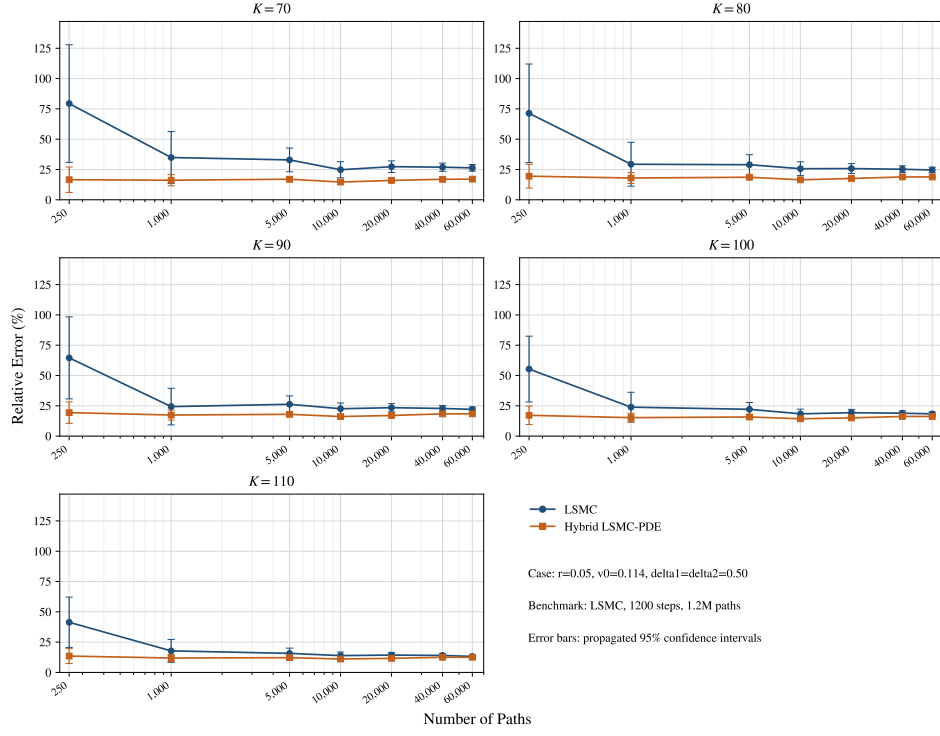


Fig. 4 Relative errors for the fixed 60-step path-sweep experiment. Error bars are induced by the 95% confidence intervals of the direct estimators.

The high-path confirmation indicates that the larger errors observed in the moderate-budget sweeps are not caused by the Bermudan stopping formulation itself. They are primarily a numerical-budget effect for this parameter set, and the Hybrid prices recover the benchmark scale once the time grid and asset grid are refined together with the path budget.

Positive-rate square-root gDMR price tables

Table 8 Benchmark prices. Prices are rounded to three decimals.

Strike	70	80	90	100	110
LSMC benchmark	3.734	5.384	7.684	10.960	15.732

Table 9 Fixed-path step-sweep prices with 20,000 paths. Prices are rounded to three decimals.

Method/step	24	48	72	96
LSMC K=70	5.268	4.853	4.564	4.458
Hybrid LSMC-PDE K=70	4.773	4.435	4.259	4.156
LSMC K=80	7.417	6.916	6.507	6.360
Hybrid LSMC-PDE K=80	6.928	6.473	6.237	6.106
LSMC K=90	10.300	9.603	9.153	8.949
Hybrid LSMC-PDE K=90	9.750	9.174	8.876	8.718
LSMC K=100	13.956	13.291	12.743	12.512
Hybrid LSMC-PDE K=100	13.514	12.825	12.471	12.291
LSMC K=110	19.020	18.244	17.642	17.351
Hybrid LSMC-PDE K=110	18.558	17.798	17.414	17.225

Table 10 Fixed-path step-sweep prices with 60,000 paths. Prices are rounded to three decimals.

Method/step	24	48	72	96
LSMC K=70	5.078	4.779	4.597	4.449
Hybrid LSMC-PDE K=70	4.805	4.482	4.282	4.145
LSMC K=80	7.229	6.812	6.559	6.379
Hybrid LSMC-PDE K=80	6.980	6.550	6.282	6.098
LSMC K=90	10.034	9.514	9.199	8.983
Hybrid LSMC-PDE K=90	9.820	9.282	8.947	8.715
LSMC K=100	13.755	13.112	12.791	12.508
Hybrid LSMC-PDE K=100	13.590	12.957	12.566	12.295
LSMC K=110	18.720	17.944	17.666	17.420
Hybrid LSMC-PDE K=110	18.625	17.935	17.520	17.235

Table 11 Fixed 48-step path-sweep prices. Prices are rounded to three decimals.

Method/strike	250	1,000	5,000	10,000	20,000	40,000	60,000
LSMC K=70	5.809	5.050	5.100	4.855	4.853	4.851	4.779
Hybrid LSMC-PDE K=70	4.537	4.550	4.442	4.371	4.435	4.463	4.482
LSMC K=80	8.252	7.169	7.234	6.909	6.916	6.894	6.812
Hybrid LSMC-PDE K=80	6.658	6.642	6.492	6.393	6.473	6.529	6.550
LSMC K=90	12.081	9.869	10.036	9.632	9.603	9.606	9.514
Hybrid LSMC-PDE K=90	9.442	9.389	9.207	9.079	9.174	9.259	9.282
LSMC K=100	16.401	13.910	13.664	13.237	13.291	13.282	13.112
Hybrid LSMC-PDE K=100	13.138	13.073	12.868	12.717	12.825	12.931	12.957
LSMC K=110	22.036	19.229	18.582	18.142	18.244	18.207	17.944
Hybrid LSMC-PDE K=110	18.116	18.092	17.843	17.690	17.798	17.909	17.935

Table 12 Fixed 60-step path-sweep prices. Prices are rounded to three decimals.

Method/strike	250	1,000	5,000	10,000	20,000	40,000	60,000
LSMC K=70	6.699	5.036	4.963	4.659	4.755	4.739	4.717
Hybrid LSMC-PDE K=70	4.353	4.337	4.368	4.280	4.331	4.365	4.370
LSMC K=80	9.226	6.963	6.942	6.765	6.772	6.741	6.709
Hybrid LSMC-PDE K=80	6.433	6.351	6.387	6.272	6.331	6.398	6.400
LSMC K=90	12.645	9.554	9.702	9.419	9.485	9.437	9.378
Hybrid LSMC-PDE K=90	9.173	9.019	9.065	8.924	8.992	9.094	9.094
LSMC K=100	17.024	13.577	13.384	12.974	13.076	13.029	12.973
Hybrid LSMC-PDE K=100	12.841	12.629	12.690	12.528	12.607	12.738	12.736
LSMC K=110	22.241	18.528	18.207	17.914	17.987	17.937	17.810
Hybrid LSMC-PDE K=110	17.855	17.602	17.649	17.475	17.560	17.699	17.700